A geometric region closed by the Bernoulli reserve

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The geometric budget and the Bernoulli reserve have the same quadratic factor $Q(p)$. We use this to close an entire part of the remaining left range. The proof keeps all phases and all admissible noise in that region. It uses no count-tail certificate: fourth moments give a summable bound for the signed curvature in the exact crossing identity.

Write $\phi_0 = (2\pi)^{-1/2}$, $\sigma = \sqrt{10}$ and

$$C_E = (3+\sigma)\phi_0/6, \quad Q(p) = (6+2\sigma)(p-p_*)^2, \quad p_* = (4-\sigma)/2.$$

For $Y = B + U$, put $p = \mathbb{P}(B=1)$, $q = 1-p$, $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $\ell = (v+s)^{-1/2}$ and $X = \ell(Y-p)$. The two conditional laws U_0, U_1 are centered. The notation D denotes the actual support-gap budget of the v13 geometric theorem:

$$D = (3+\sigma)\mathbb{E}|X|^3 - \mathbb{E}X^3 - 3\sum_{(a,b)} \tau_{a,b}(b-a)^2.$$

Theorem 1 (An excluded probability–index region)**.** *Let $n \geq 42$ be an integer and $1/3 \leq p \leq 2/5$. Suppose*

$$-7/10 \leq U_0 \leq a_0, \qquad -a_1 \leq U_1 \leq 7/10, \quad a_0, a_1 \geq 0, \quad a_0 + a_1 \leq 17/250,$$

$$0 \leq \lambda \leq 8/25, \qquad nD \leq 129/4, \qquad |H| \leq 46/25.$$

If $nQ(p)/\sqrt{v} \geq 24$, then, for the sum S_n of n iid copies of X,

$$\sqrt{n}\{\mathbb{P}(S_n \leq H) - \Phi(H/\sqrt{n})\} - C_E\mathbb{E}|X|^3 < -\frac{1}{10n}.$$

The acquired budget $nD \leq 32.20665378$ is smaller than $129/4$; the cap $\lambda < 8/25$ on this probability range is already proved in v13. Thus an acquired witness with nonnegative signed excess must satisfy

$$nQ(p)/\sqrt{pq} < 24. \tag{1}$$

This improves the previous necessary bound $nQ(p)/\sqrt{pq} \leq 32.20665378$ by using the actual CDF, rather than by slightly changing a noise ceiling. The finite base and the remaining region below (1) are not settled. No quantitative Bernoulli reserve is asserted below $p = 1/3$.

1 A summable bound for the signed curvature

Let W_j be a sum of $n-j$ independent copies of U_0 and j of U_1. Set $\pi_j = \mathbb{P}(\mathrm{Bin}(n,p) = j)$ and

$$B_j(t) = \mathbb{E}(t-j-W_j)_+ - (t-j)_+, \qquad \mathcal{C}(t) = \sum_{j=1}^{n-1}(\pi_{j+1} - \pi_j)B_j(t).$$

The following lemma is independent of the geometric budget and of the Bernoulli theorem. It is useful on either probability range.

Lemma 2. *Let $n \geq 2$, $0 < p < 1$, and put $c = (np - q - t)_+ \leq 1$. Suppose $T, M \geq 0$ satisfy $\mathbb{E}W_j^2 \leq T$ and $\mathbb{E}W_j^4 \leq M$ for every j with $\pi_{j+1} > \pi_j$. If*

$$\max_r \mathbb{P}(\mathrm{Bin}(n+1,p) = r) \leq \frac{\beta}{\sqrt{(n+1)v}},$$

then

$$\sqrt{n}\,\mathcal{C}(t) \leq \frac{\beta}{nv^{3/2}} \left\{ \frac{T}{4} + c\sqrt{T} + \frac{27M}{256}\left(\frac{5}{3} + \frac{5c}{4}\right)\right\}. \tag{2}$$

For bounded conditional laws $|U_j| \leq A$, one may take $T = \lambda/q$ and $M = A^2T + 3T^2$.

Proof. For $a = t - j$, centering gives

$$B_j = \tfrac{1}{2}\{\mathbb{E}|a - W_j| - |a|\} \geq 0, \qquad B_j \leq \sqrt{T}/2, \qquad a_+ B_j \leq T/4.$$

The last bound follows from $\sqrt{a^2 + T} - a \leq T/(2a)$ for $a > 0$. Also $(x - a)_+ \leq 27x^4/(256a^3)$ for $a > 0$: the maximum of $(x - a)/x^4$ is attained at $x = 4a/3$. Consequently

$$B_j \leq 27M/(256|a|^3) \quad (a \neq 0).$$

The exact adjacent-mass identity, with $\widetilde{\pi}_r = \mathbb{P}(\mathrm{Bin}(n+1,p) = r)$, is

$$\pi_{j+1} - \pi_j = \frac{np - q - j}{(n+1)v}\widetilde{\pi}_{j+1}.$$

Discard only its nonpositive terms. Its positive numerator is at most $(a + c)_+$. Among the remaining integer mesh points there is at most one with $-c < a < 0$ and at most one with $0 \leq a < 1$. Their total is bounded by $c\sqrt{T} + T/4$. For $a \geq 1$, list the points as $a = b + i$, $i \geq 1$, with $0 \leq b < 1$. The fourth-moment bound and

$$\sum_{i \geq 1} i^{-2} < 5/3, \qquad \sum_{i \geq 1} i^{-3} < 5/4$$

give the last term in braces. Both series bounds follow from their first few terms and an integral remainder. Multiply by the mass bound and use $\sqrt{n}/(n+1)^{3/2} \leq 1/n$.

Finally, a positive mass difference requires $j < np - q < np$. Writing $s_i = \mathbb{E}U_i^2$, the variance $(n - j)s_0 + js_1$ is at most λ/q, because its coefficients relative to nqs_0 and nps_1 are at most $1/q$ and 1, respectively. Independence gives $\mathbb{E}W_j^4 = \sum \mathbb{E}U_i^4 + 3(\mathbb{E}W_j^2)^2 - 3\sum(\mathbb{E}U_i^2)^2 \leq A^2T + 3T^2$. The moment and count allocations remain those of the same two conditional laws. □

For $p \in [1/3, 2/5]$ and $n \geq 42$ one may take $\beta = 41/100$. Here is a short finite justification. Put $N = n + 1 \geq 43$, and let $r = m/N$ for a binomial mode $m = \lfloor (N+1)p \rfloor$. Then $-q \leq m - Np \leq p$ and

$$r(1 - r) \geq v\{1 - 1/N - 2/N^2\} > 39v/40.$$

Finite Stirling, after discarding the nonpositive entropy exponent and negative denominator remainders, bounds the maximum mass by

$$\frac{\phi_0}{\sqrt{Nr(1-r)}} e^{1/(12N)} < \frac{(2/5)\sqrt{40/39}(516/515)}{\sqrt{Nv}} < \frac{41/100}{\sqrt{Nv}}.$$

The final comparison is checked by squaring positive rational quantities.

2 The geometric cost leaves less than .09 noise

The v13 inward-cubic inequality, with $\bar{d} = 17/250$, gives

$$nD \geq \frac{nQ/\sqrt{v} + \lambda \min(Y_0, Y_1)/v^{3/2}}{(1 + \lambda/(nv))^{3/2}},$$

$$Y_0 = 3(\sigma+3)p - (\sigma+7)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma+1)p},$$

$$Y_1 = 3(\sigma+1)q - (\sigma+5)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma-1)q}.$$

On $[1/3, 2/5]$, both $Y_j/v^{3/2} > 100$. To see this without a grid, divide the positive terms by $v^{3/2}$. For Y_0 the two denominators are $\sqrt{p}\,q^{3/2}$ and $pq^{3/2}$; for Y_1 they are $p^{3/2}\sqrt{q}$ and $p^{3/2}q$. Their respective maxima on this interval are smaller than

$$63/200, \quad 93/500, \quad 49/250, \quad 19/125.$$

Differentiating their squares proves the endpoint locations (the second has its maximum at $p = 2/5$). Also $v^{3/2} > 13/125$, $\sqrt{6(1-\bar{d})(\sigma+1)} > 24/5$, and $\sqrt{6(1-\bar{d})(\sigma-1)} > 69/20$. Replacing σ by 3.162277660 in positive terms and by 3.162277661 in negative terms now proves both bounds by rational arithmetic.

Since $v \geq 2/9$, $n \geq 42$ and $\lambda \leq 8/25$, one has $x = \lambda/(nv) \leq 6/175$. The elementary inequality $(1+x)^{3/2} \leq 1 + 3x/2 + 3x^2/8$ implies $(1+x)^{3/2} \leq 1 + (31/20)x$. Hence the hypotheses of Theorem 1 yield

$$24 + 100\lambda \leq \frac{129}{4}\left(1 + \frac{31}{20}\frac{3\lambda}{28}\right), \qquad \text{and therefore} \qquad \lambda < 9/100.$$

There has been no optimization over independent moment or noise parameters.

3 Interpolation preserves the Bernoulli reserve

Put $t = np + H\sqrt{v+s}$, $k = \lfloor t \rfloor$, and $u = t - k \in [0, 1)$. The support assumptions give the crossing theorem of v13, including $na_0 \leq t < n(1 - a_1)$. Its exact weighted identity implies

$$\mathbb{P}(S_n \leq H) \leq (1-u)F_B(k) + uF_B(k+1) + \mathcal{C}(t). \tag{3}$$

The nonnegative weighted crossing loss was discarded, not estimated by an unrelated law. The identity uses closed lower crossing and strict upper crossing events and remains valid at every atom.

Schulz's theorem, in its published $p \in [1/3, 2/3]$ range, gives for every real argument the strict Bernoulli reserve

$$\sqrt{n}\{F_B(x) - \Phi((x-np)/\sqrt{nv})\} - C_E\rho_B < -\frac{\phi_0 Q(p)}{6\sqrt{v}} \leq -\frac{4\phi_0}{n}, \qquad \rho_B = (1-2v)/\sqrt{v}.$$

Only the two integer arguments $k, k+1$ are used in (3). This formula is not extended below $p = 1/3$.

We bound the normal interpolation error with its sign. Write $w = t - np$, and $G(x) = \Phi((x-np)/\sqrt{nv})$. The interpolation remainder satisfies

$$\sqrt{n}\{(1-u)G(k) + uG(k+1) - G(t)\} \leq \frac{\phi_0}{2nv^{3/2}}u(1-u)(u-w)_+.$$

Indeed $G''(x) \le \phi_0(np-x)_+/(nv)^{3/2}$ on the entire segment. Since $v+s<1/4$, one has $|w|<23/25$ and $c=(np-q-t)_+<8/25$. For $H\le 0$, the normal change from variance v to $v+s$ is favorable. The interpolation error is bounded by

$$\frac{2/5}{nv^{3/2}}\left(\frac{23}{200}+\frac{2}{27}\right),$$

using $u(1-u)\le 1/4$ and $u^2(1-u)\le 4/27$. For $H\ge 0$ the interpolation error is at most $(2/5)(2/27)/(nv^{3/2})$, and the variance repair is at most

$$\frac{\phi_0 ws}{2v^{3/2}}<\frac{(2/5)(23/25)(9/100)/2}{nv^{3/2}}.$$

This uses $1-(1+x)^{-1/2}\le x/2$; here $c=0$.

The third-moment correction is favorable throughout this theorem. The actual marginal expansion and inward cubic bounds give

$$(v+s)^{3/2}(\rho-\rho_B)\ge s\left\{3p-\frac{3}{2}(1-2v)-\frac{17}{250}-\frac{3(1-2v)s}{8v}\right\}>0\quad(s>0).$$

Indeed $p\ge 1/3$, $1-2v\le 5/9$, $v\ge 2/9$ and $s<9/(100\cdot 42)$ make the braces positive. At $s=0$ the correction is exactly zero. This favorable sign is not asserted on the whole left range below $1/3$.

Finally Lemma 2 applies with

$$T=3/20,\quad M=141/1000,\quad \beta=41/100,\quad \sqrt{T}<97/250,\quad v^{3/2}>13/125.$$

Let

$$S_c=\frac{3}{80}+c\frac{97}{250}+\frac{27}{256}\frac{141}{1000}\left(\frac{5}{3}+\frac{5c}{4}\right).$$

For $H\le 0$ and $H\ge 0$, respectively, the total unfavorable errors in (3), multiplied by n, are strictly smaller than

$$E_-:=\frac{(41/100)S_{8/25}+(2/5)(23/200+2/27)}{13/125},$$

$$E_+:=\frac{(41/100)S_0+(2/5)(2/27)+(2/5)(23/25)(9/100)/2}{13/125}.$$

Exact rational arithmetic gives

$$E_-,E_+<4\frac{19943}{50000}-\frac{1}{10}<4\phi_0-\frac{1}{10}.$$

Combine this with the strict Bernoulli reserve to prove Theorem 1. Both signs of H, every phase $u\in[0,1)$, zero noise and all threshold atoms have been included.

Dependencies and limits

The geometric inequality is proved in `mathematics/simplification-v13/2026-10-28/geometry-noise.tex`; the exact weighted crossing identity is in the sibling `crossing.tex`. The external Bernoulli theorem is Theorem 1, printed page 1, of Schulz's *The optimal Berry–Esseen constant in the binomial case* (2016), original thesis. Finite Stirling is the same explicit remainder bound used in the retained count normalization. No Fourier, Poisson or count-generation program is an input to the new finite arithmetic.

For acquired witnesses the actual central gap, the coarse geometric budget, left support strips and raw threshold still require their existing contact, sharp-predecessor and sign-specific acquisition. Applying the statement to a reflected branch requires these inputs for the reflected law. No global attainment, independent finite base through 41, comparison below $p=1/3$, or full conjecture is proved here.